Muntaha Kamal

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EXPERIENCE

Tata Electronics GIFT City, Ahmedabad, Gujarat

Assistant Manager – AI Engineer *December 2025 – Present*

- Owned end-to-end development of a **multimodal GraphRAG system** that transforms semiconductor technical documents into department-specific learning modules using **Neo4j** knowledge graphs and vector retrieval. Evaluated alternative approaches and built custom ingestion with controlled image and table cropping.
- Scoped GraphRAG retrieval by filtering chunk, entity, and relationship vector candidates **before top-k selection**; preserved source-specific graph claims and image provenance when documents shared entities. Validated selected-document evidence through regression tests and persisted-store checks.
- Completed a **five-month Taiwan assignment** as the AI specialist, documenting fab systems, wafer movement, and production data flows to build domain expertise and strengthen collaboration with IT, module, and fab-application teams.

Micron Technology Hyderabad, Telangana

Data Scientist II *January 2024 – November 2025*

- Owned NAND reliability modeling with **Lasso and XGBoost**, comparing direct die-level models with hierarchical wafer-baseline and die-residual models using wafer-relative features. Predictions supported the core team's re-ratio optimization, contributing to an approximately **15–20% relative increase** in the share of dies qualifying for eTLC.
- Contributed ML models using time-zero test metrics to a separate cross-functional effort identifying dies eligible for earlier qualification; presented at Micron's internal NAND Summit.
- Developed **NAND FDV anomaly analysis** in Python/BigQuery using rolling-median baselines, Savitzky–Golay smoothing, and positive-residual thresholds to flag local error-count variation below static ECC limits. Prioritized recurring outliers for interactive Plotly investigation; first-authored an **internal Micron TLP technical paper**.
- Co-developed the **Unified Reliability Modeling Platform**, contributing to scoping and owning reusable, tested Python modeling-workflow modules, visualization, and evaluation to support consistent reliability studies. Applied rigorous peer review to code and design decisions.

Data Scientist I *June 2022 – December 2023*

- Trained a **Random Forest classifier** on historical backend engineering-test data and engineer decisions to recommend retaining or downgrading material grade for Quality Assurance Work Requests (QAWR). Provided feature/deviation explanations and low-confidence abstention for engineer review, reducing turnaround by approximately **14 days**.
- Co-developed a **GCP Airflow ETL and model-inference workflow** with failure alerts, reducing combined model-development and data-engineering effort by approximately **50%**.
- Built a separate twice-weekly pass-notification workflow using **Cloud Scheduler, Pub/Sub, and Cloud Functions** to check QAWR material against pass criteria and email alerts when it qualified.

SELECTED PROJECT

CommitEcho | *Python, MCP, SQLite, Git* **September – October 2026**

- Built a **developer tool** integrating a local **Model Context Protocol (MCP) server, CLI, and agent skills** to automatically capture engineering decisions and alternatives in configured coding-agent workflows and recall their rationale on request.
- Implemented Git-linked, portable decision records with code-change fingerprint verification and revision-scoped **SQLite FTS5** retrieval across sessions and clones. Published **v0.2.0 on PyPI** with Windows/Linux **GitHub Actions CI**. Source: github.com/Muntaha15/CommitEcho.

TECHNICAL SKILLS

Languages	: Python, SQL, C++, Java, TypeScript
Machine Learning	: PyTorch, scikit-learn, XGBoost, NumPy, pandas
Generative AI	: RAG, multimodal GraphRAG, embeddings, AI agents and workflows, MCP
Graph & Vector DBs	: Neo4j, NanoVectorDB
Cloud & Data	: GCP, BigQuery, Airflow, Pub/Sub, Cloud Functions, Cloud Scheduler
Engineering & Tools	: Docker, FastAPI, pytest, Linux, Git, GitHub Actions, Plotly

EDUCATION

Indian Institute of Technology, Delhi **July 2018 – June 2022**

B.Tech in Production & Industrial Engineering *CGPA: 7.97/10*